1. Justification for Using a Hub-and-Spoke Architecture

Justification for the Hub-and-Spoke Architecture for MediLink Health Network

MediLink Health Network operates four clinics located in Kampala, Mukono, Mbarara, and Gulu. Each clinic maintains its own records, resulting in challenges such as duplicate patient identities, inconsistent data, and difficulties consolidating information for management and Ministry of Health reporting. To address these challenges, the proposed Enterprise Data Management (EDM) platform will adopt a **hub-and-spoke architecture** that centralizes data integration and management while allowing individual clinics to maintain their local operations.

The hub-and-spoke architecture consists of a central hub connected to multiple spokes. In the MediLink implementation, the four clinics represent the spokes, while the central data management platform represents the hub. Each clinic acts as a source of patient, visit, clinical, and billing information. These records are transferred to the central platform, where they undergo validation, standardization, data quality checks, and patient identity resolution before being consolidated into trusted records.

A major justification for this architecture is the establishment of a **single, trusted patient record** across the health network. Currently, the same patient may have different records at different clinics. Through Master Data Management (MDM), the central hub can identify records belonging to the same patient and link them to a golden record. This reduces duplication and improves consistency without requiring every clinic to independently maintain a separate master patient database.

Secondly, the architecture supports improved data quality and governance. The central hub provides a common location for applying standardized validation rules, monitoring data quality, maintaining metadata, and enforcing data ownership policies. For example, patient telephone numbers can be standardized into the required format, while conflicting identity information can be referred to a data steward for review rather than merged automatically. This creates a consistent approach to managing information across all four clinics.

Thirdly, hub-and-spoke architecture simplifies reporting and decision-making. Instead of manually collecting spreadsheets and reports from each clinic, the central platform can consolidate the required information for analysis. A data warehouse connected to the hub can support reporting on outpatient attendance, common diagnoses, and revenue by service. This can reduce the time and effort required to prepare monthly Ministry of Health reports.

The architecture also supports stronger security and accountability. Centralized access controls, auditing, and data classification can help ensure that staff access only the information required

for their responsibilities. Sensitive patient information can be protected through appropriate permissions, masking, and controlled access to the master records and analytical datasets.

Another advantage is maintainability. Common data standards and integration processes can be managed centrally rather than implemented differently at every clinic. Additional clinics could also be connected as new spokes without redesigning the entire platform. However, the central hub must be protected against unauthorized access, data loss, and service interruption because it becomes a critical component of the network. Backup, recovery, and appropriate synchronization arrangements will therefore be necessary.

In conclusion, the hub-and-spoke architecture is suitable for MediLink because it provides a structured approach to integrating information from four clinics, improving patient identity management, enforcing common data quality rules, and producing consolidated reports. It supports the development of a trusted enterprise data platform while allowing the clinics to remain the originating sources of their operational information.